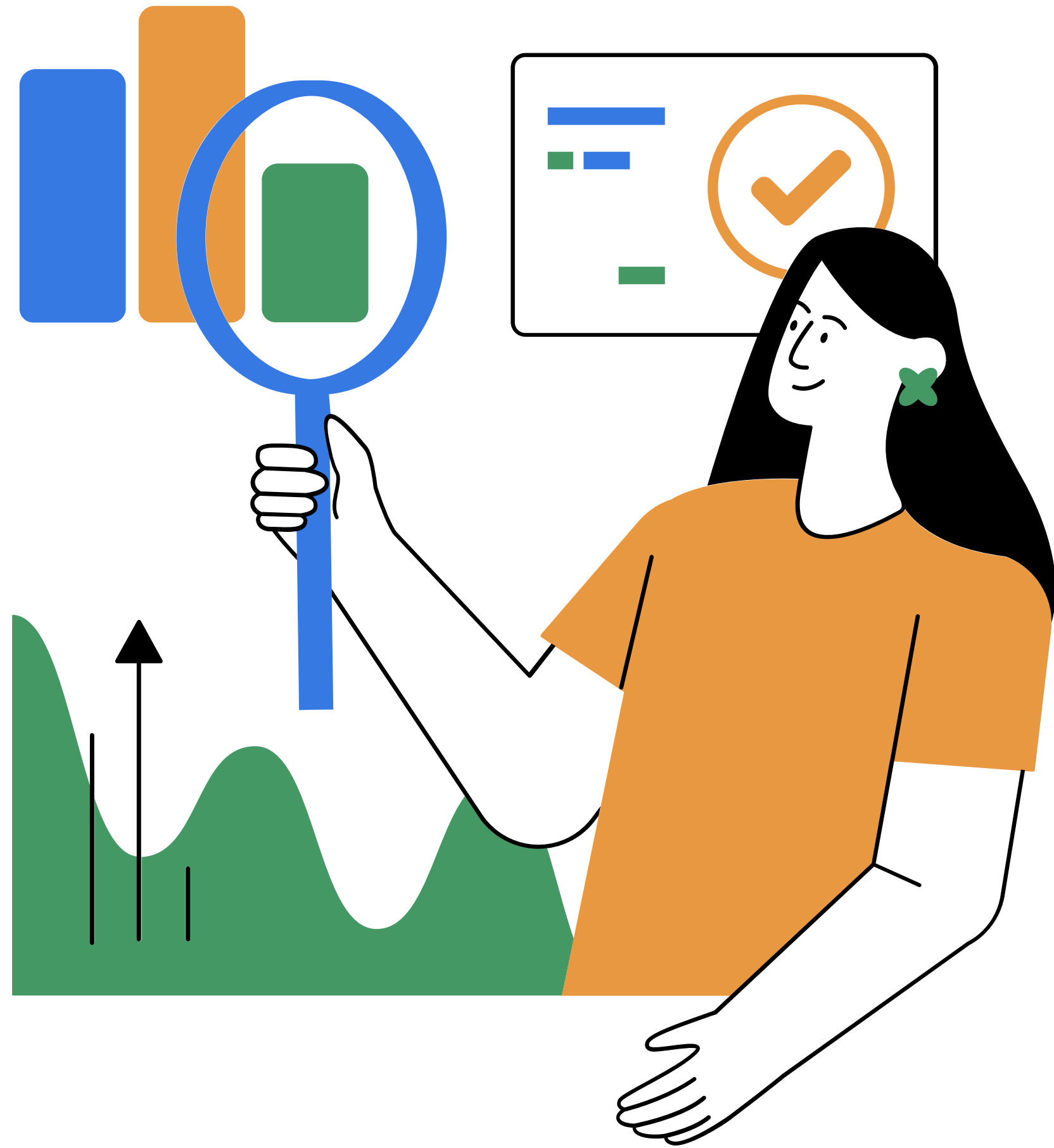


RETAINING GROWTH: REVERSING CHURN TO PROTECT REVENUE



~ANKITA SARKAR



Customer Churn KPI Summary:-

```
total_customers = len(df)
churned_customers = (df['Churn'] == 'Yes').sum()
churn_rate = (churned_customers / total_customers) * 100
avg_monthly_charge = df['MonthlyCharges'].mean()

print(f"=== KPI SUMMARY ===")
print(f"Total Customers: {total_customers:,}")
print(f"Churned Customers: {churned_customers:,}")
print(f"Overall Churn Rate: {churn_rate}%")
print(f"Average Monthly Charge: ${avg_monthly_charge:.2f}")
```

```
=== KPI SUMMARY ===
Total Customers: 7,043
Churned Customers: 1,869
Overall Churn Rate: 26.536987079369588%
Average Monthly Charge: $64.76
```

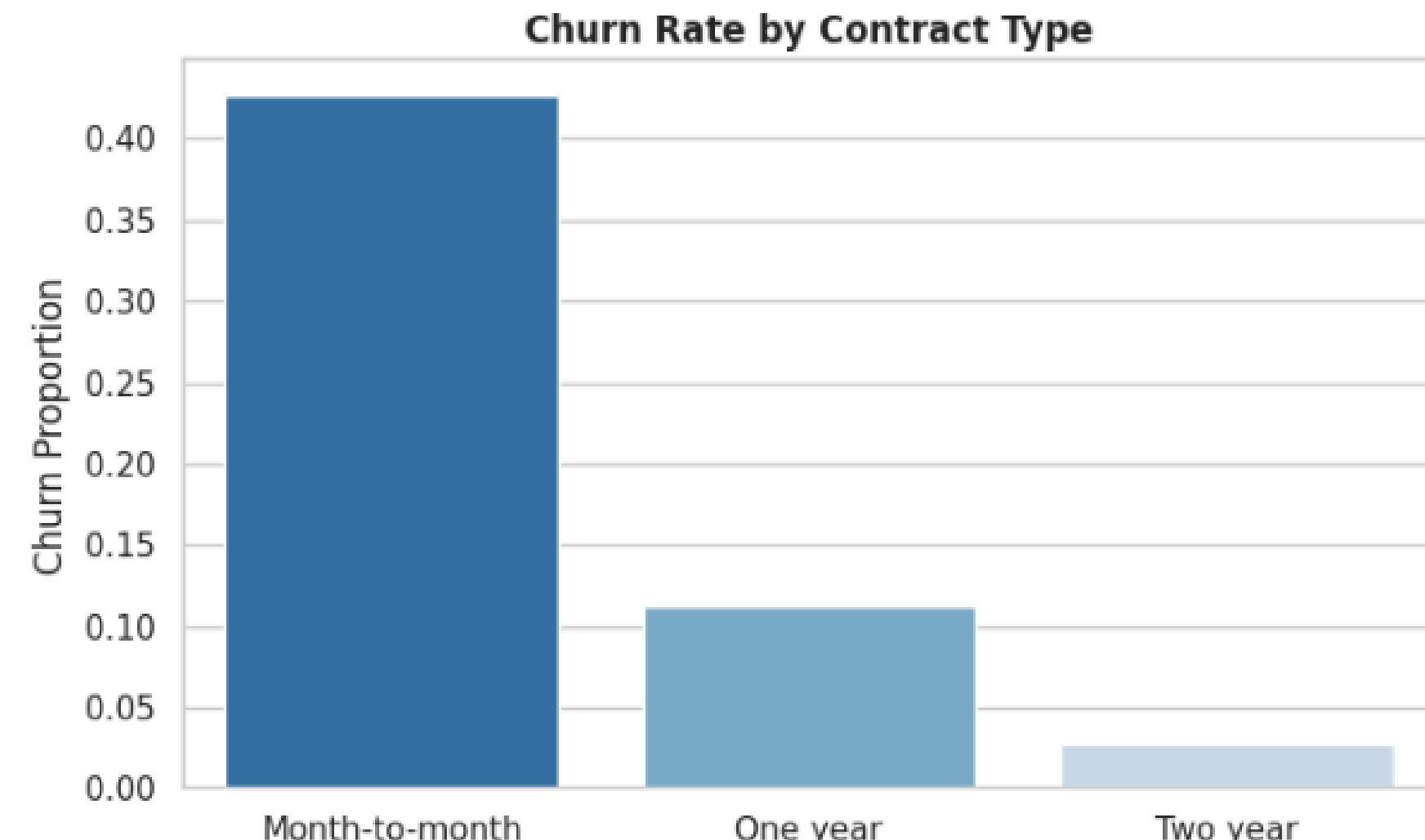
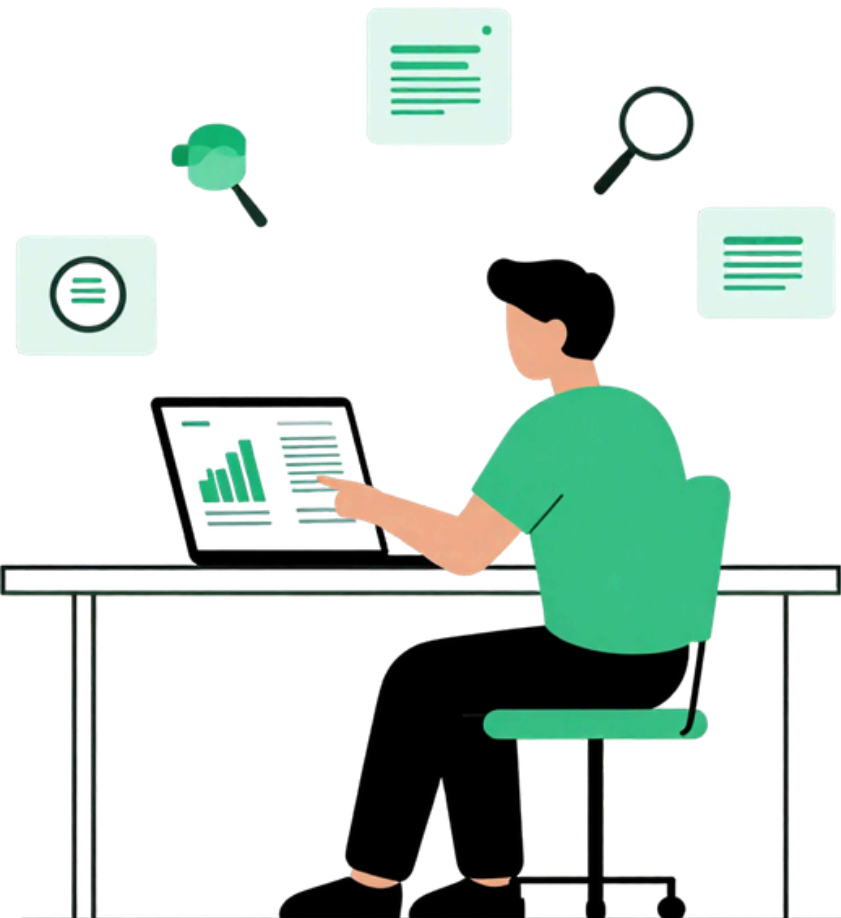


Churn Insights by Contract Type:-

```
# Set global style
sns.set_theme(style="whitegrid")
plt.rcParams.update({'font.sans-serif': 'DejaVu Sans', 'font.size': 10})

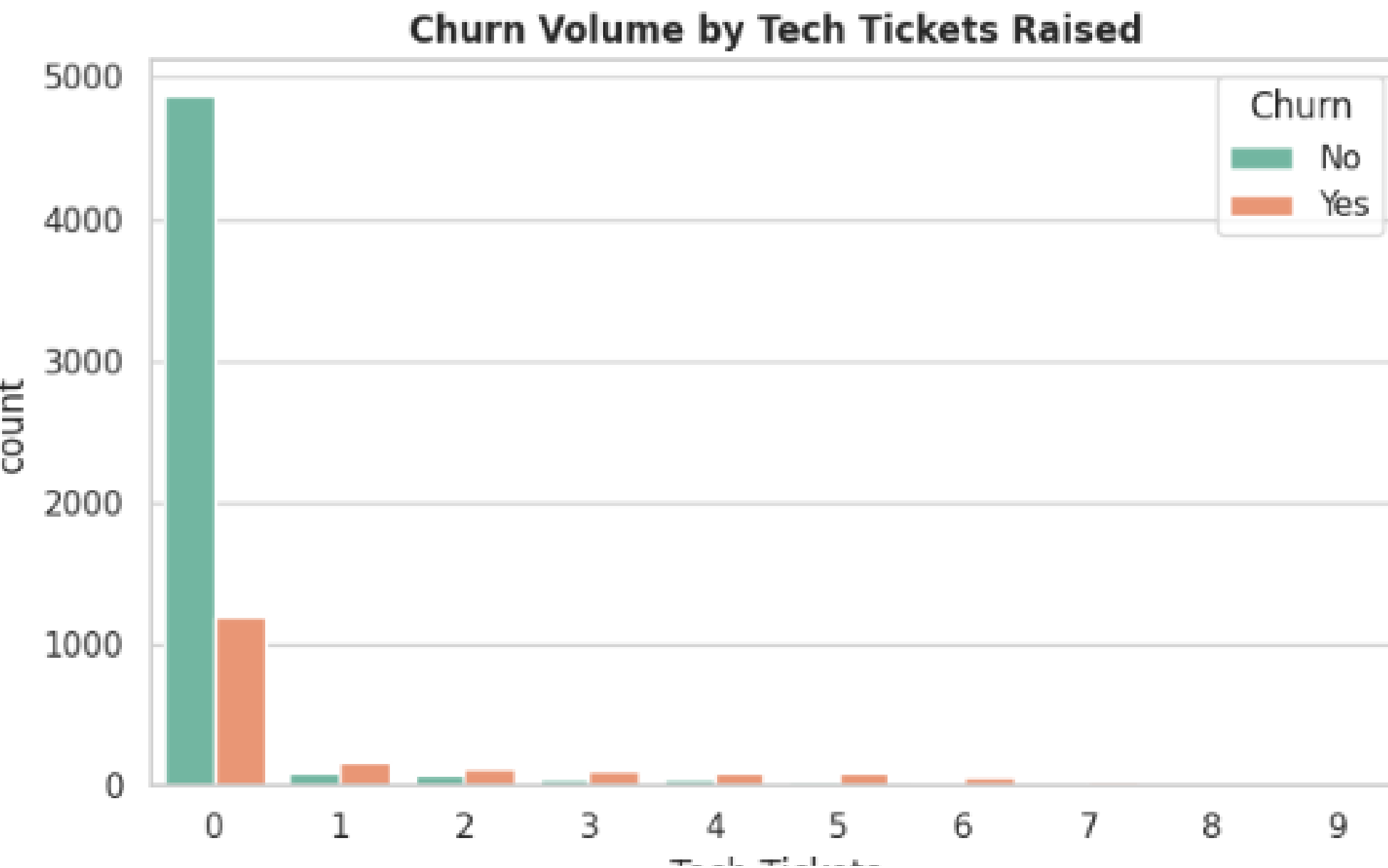
# Create 2x2 subplot canvas
fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle('Customer Churn Analysis Dashboard', fontsize=18, fontweight='bold', y=0.98)

# Panel 1: Churn Rate by Contract Type
sns.barplot(data=df, x='Contract', y='Churn_Numeric', ax=axes[0, 0], palette='Blues_r', errorbar=None)
axes[0, 0].set_title('Churn Rate by Contract Type', fontweight='bold')
axes[0, 0].set_ylabel('Churn Proportion')
```



Impact of Support Requests on Churn

```
# Panel 2: Tech Support Tickets vs Churn
sns.countplot(data=df, x='numTechTickets', hue='Churn', ax=axes[0, 1], palette='Set2')
axes[0, 1].set_title('Churn Volume by Tech Tickets Raised', fontweight='bold')
axes[0, 1].set_xlabel('Tech Tickets')
```



Customer Retention Risk Factors

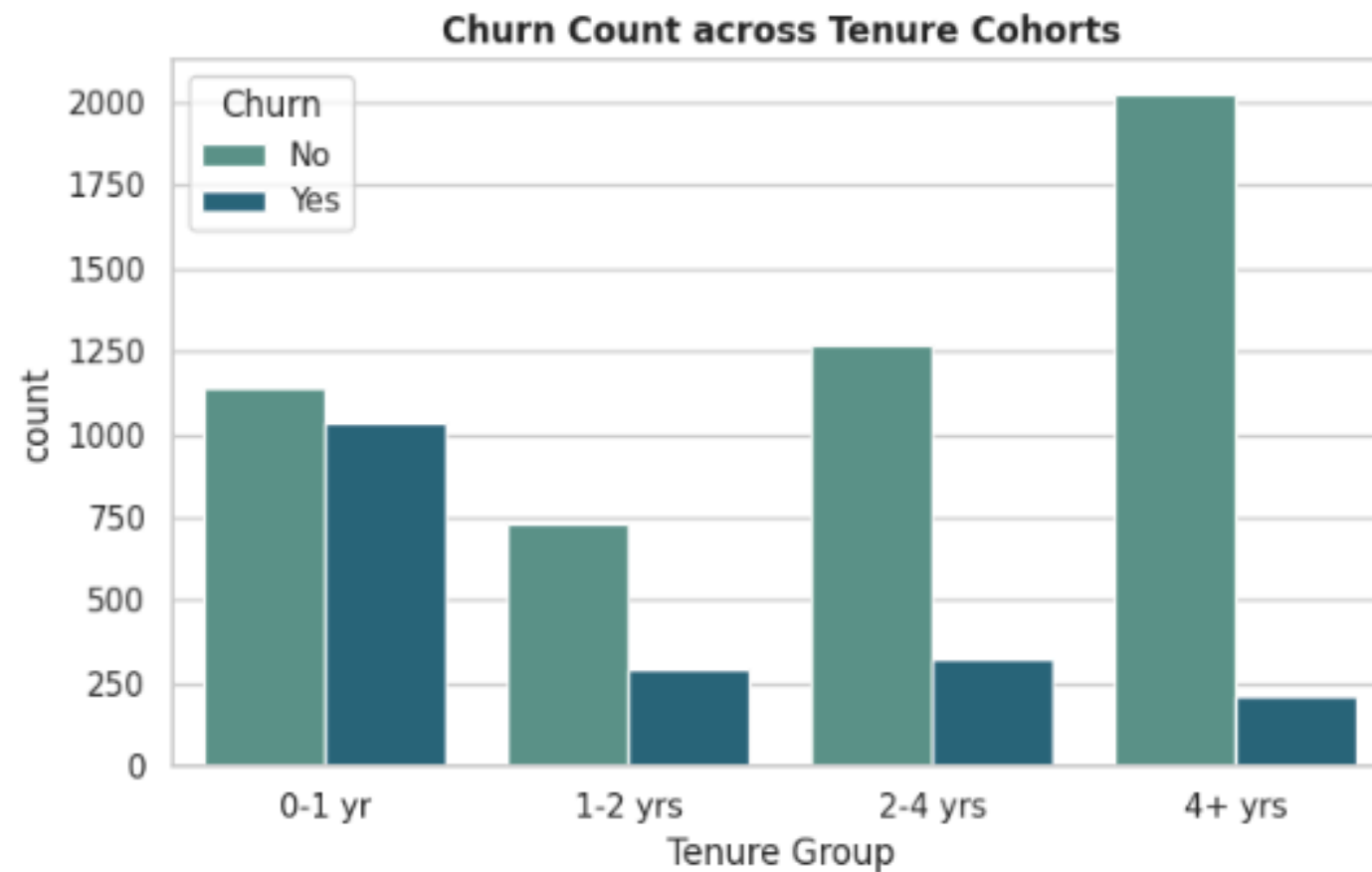
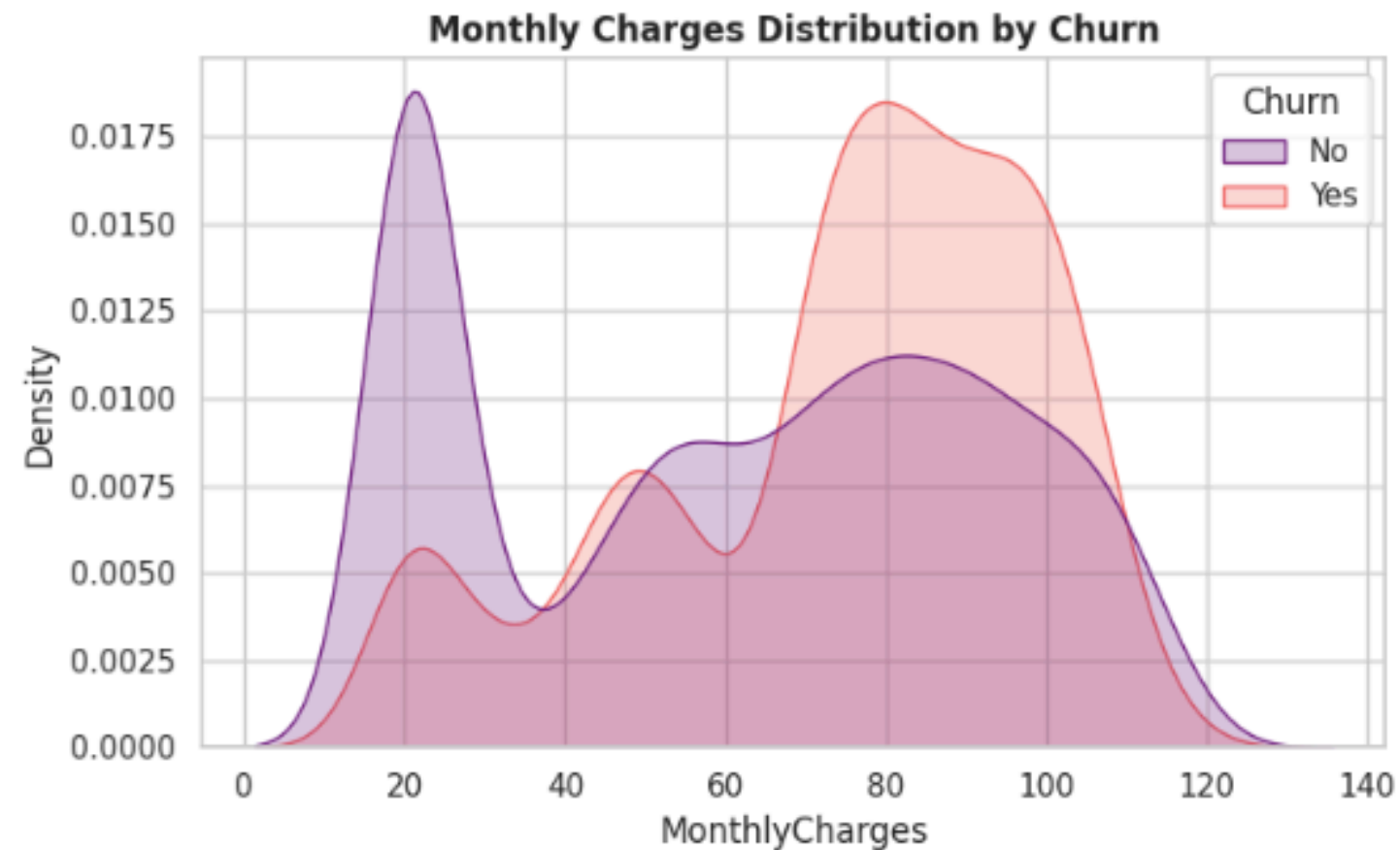
```
# Panel 3: Monthly Charge Distribution
```

```
sns.kdeplot(data=df, x='MonthlyCharges', hue='Churn', fill=True, common_norm=False, ax=axes[1, 0], palette='magma')  
axes[1, 0].set_title('Monthly Charges Distribution by Churn', fontweight='bold')
```

```
# Panel 4: Tenure Group Distribution
```

```
sns.countplot(data=df, x='Tenure_Group', hue='Churn', ax=axes[1, 1], palette='crest')  
axes[1, 1].set_title('Churn Count across Tenure Cohorts', fontweight='bold')  
axes[1, 1].set_xlabel('Tenure Group')
```

```
plt.tight_layout(rect=[0, 0.03, 1, 0.95])  
plt.show()
```



Customer Retention Risk Factors

```
# Data Prep
```

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'].replace(" ", np.nan), errors='coerce')  
df['TotalCharges'].fillna(df['TotalCharges'].median(), inplace=True)  
df['Churn_Numeric'] = df['Churn'].map({'Yes': 1, 'No': 0})
```

```
# Set Aesthetics
```

```
sns.set_theme(style="whitegrid")  
plt.rcParams.update({'font.sans-serif': 'DejaVu Sans', 'font.size': 10})
```

```
# 2. EDA Plot 1: Key Categorical Churn Drivers
```

```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))  
fig.suptitle('Exploratory Data Analysis: Key Categorical Drivers', fontsize=16, fontweight='bold')
```

Exploratory Data Analysis: Churn Drivers

```
# Demographics (Senior Citizen)
sns.barplot(data=df, x='SeniorCitizen', y='Churn_Numeric', ax=axes[0, 0], palette='Blues_d', errorbar=None)
axes[0, 0].set_title('Churn Rate by Senior Citizen Status', fontweight='bold')
axes[0, 0].set_xticklabels(['Non-Senior', 'Senior Citizen'])
axes[0, 0].set_ylabel('Churn Rate')

# Partner & Dependents
sns.barplot(data=df, x='Partner', y='Churn_Numeric', hue='Dependents', ax=axes[0, 1], palette='Set2', errorbar=None)
axes[0, 1].set_title('Churn Rate by Partner & Dependents Status', fontweight='bold')

# Internet Service Type
sns.barplot(data=df, x='InternetService', y='Churn_Numeric', ax=axes[1, 0], palette='Reds_d', errorbar=None)
axes[1, 0].set_title('Churn Rate by Internet Service Type', fontweight='bold')

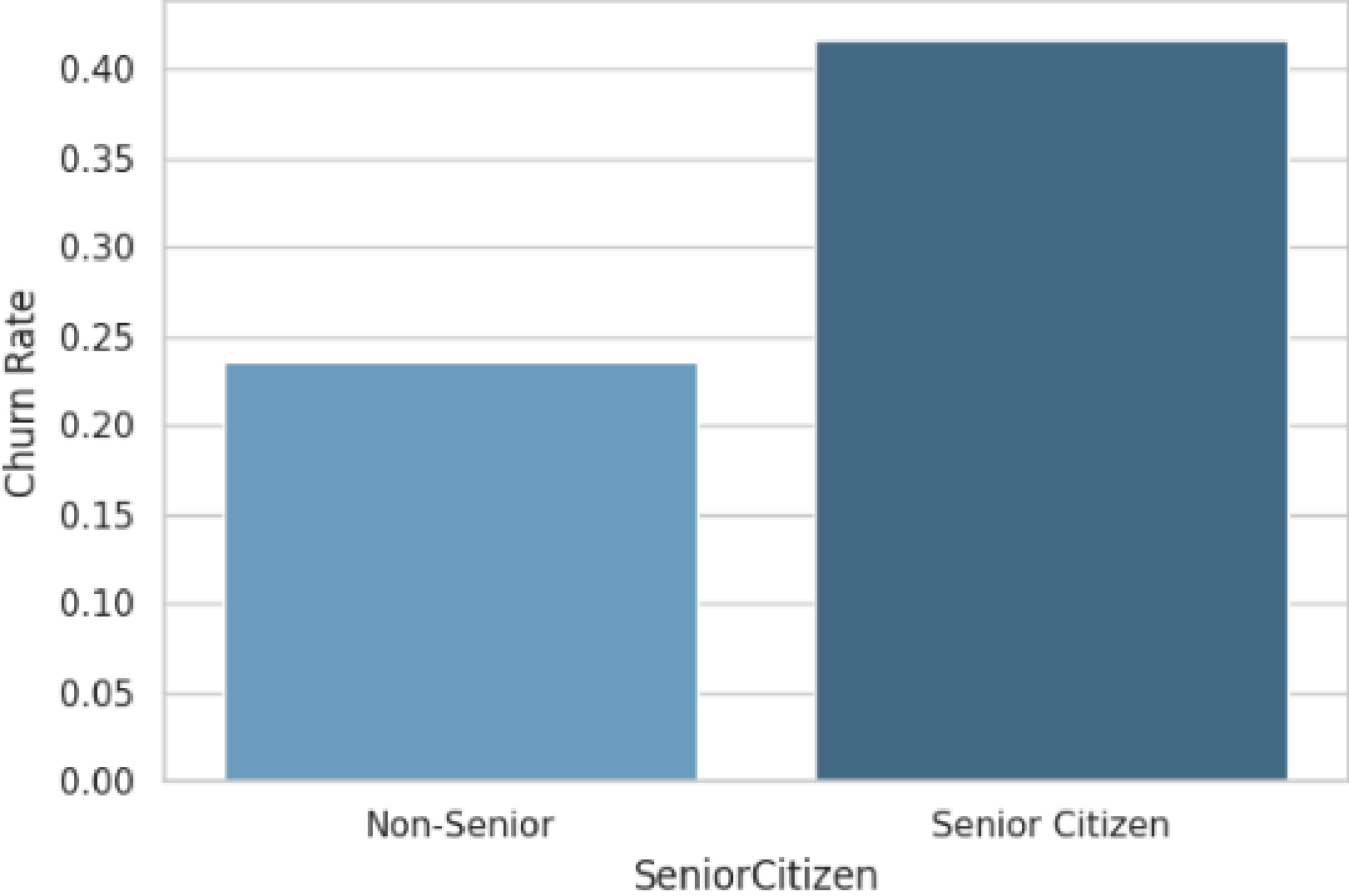
# Payment Method
sns.barplot(data=df, y='PaymentMethod', x='Churn_Numeric', ax=axes[1, 1], palette='PuBu_d', errorbar=None)
axes[1, 1].set_title('Churn Rate by Payment Method', fontweight='bold')

plt.tight_layout(rect=[0, 0.03, 1, 0.95])
plt.show()

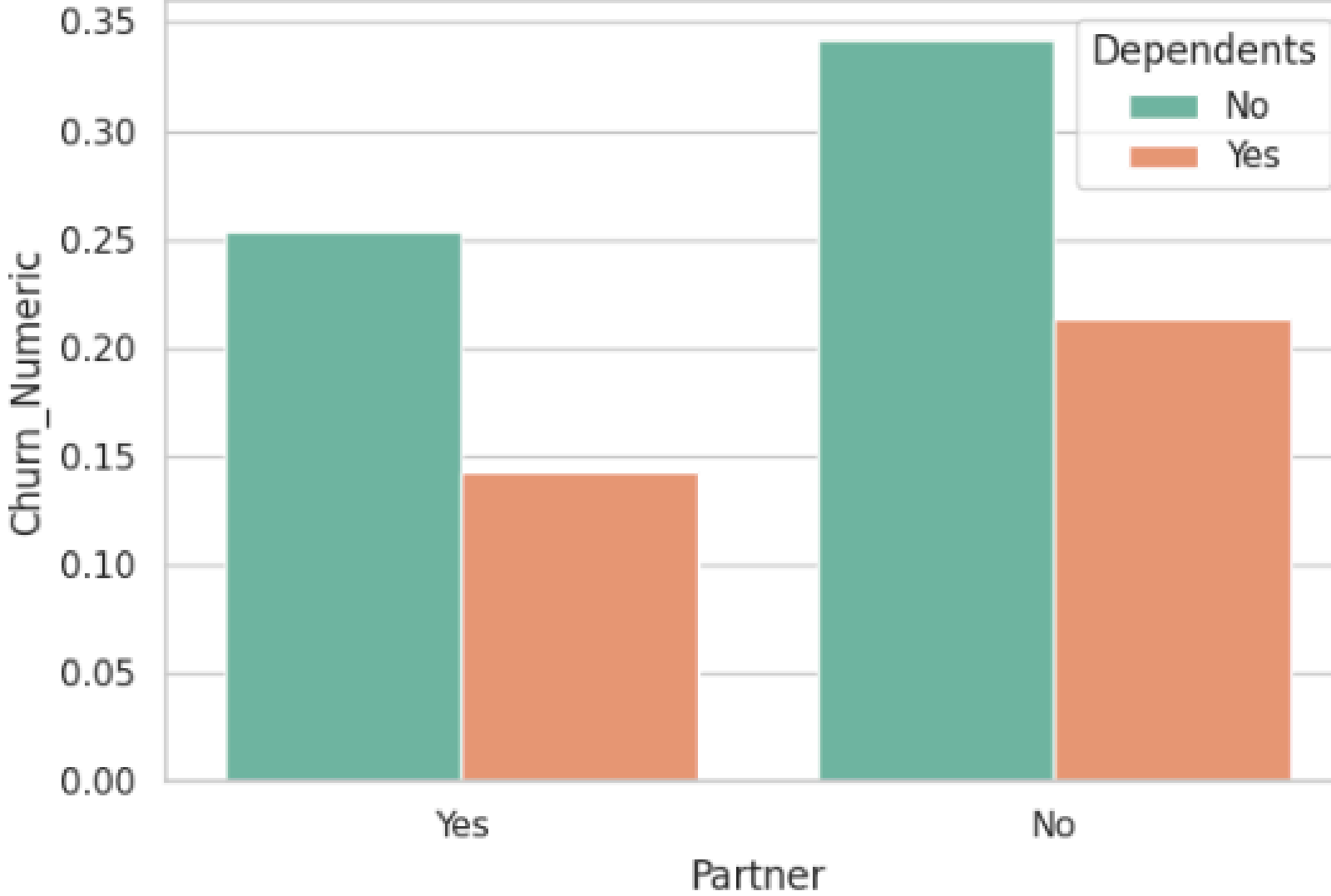
# 3. EDA Plot 2: Numerical Correlation Heatmap
plt.figure(figsize=(8, 6))
num_cols = ['tenure', 'MonthlyCharges', 'TotalCharges', 'numAdminTickets', 'numTechTickets', 'Churn_Numeric']
corr_matrix = df[num_cols].corr()
```

Exploratory Data Analysis: Key Categorical Drivers

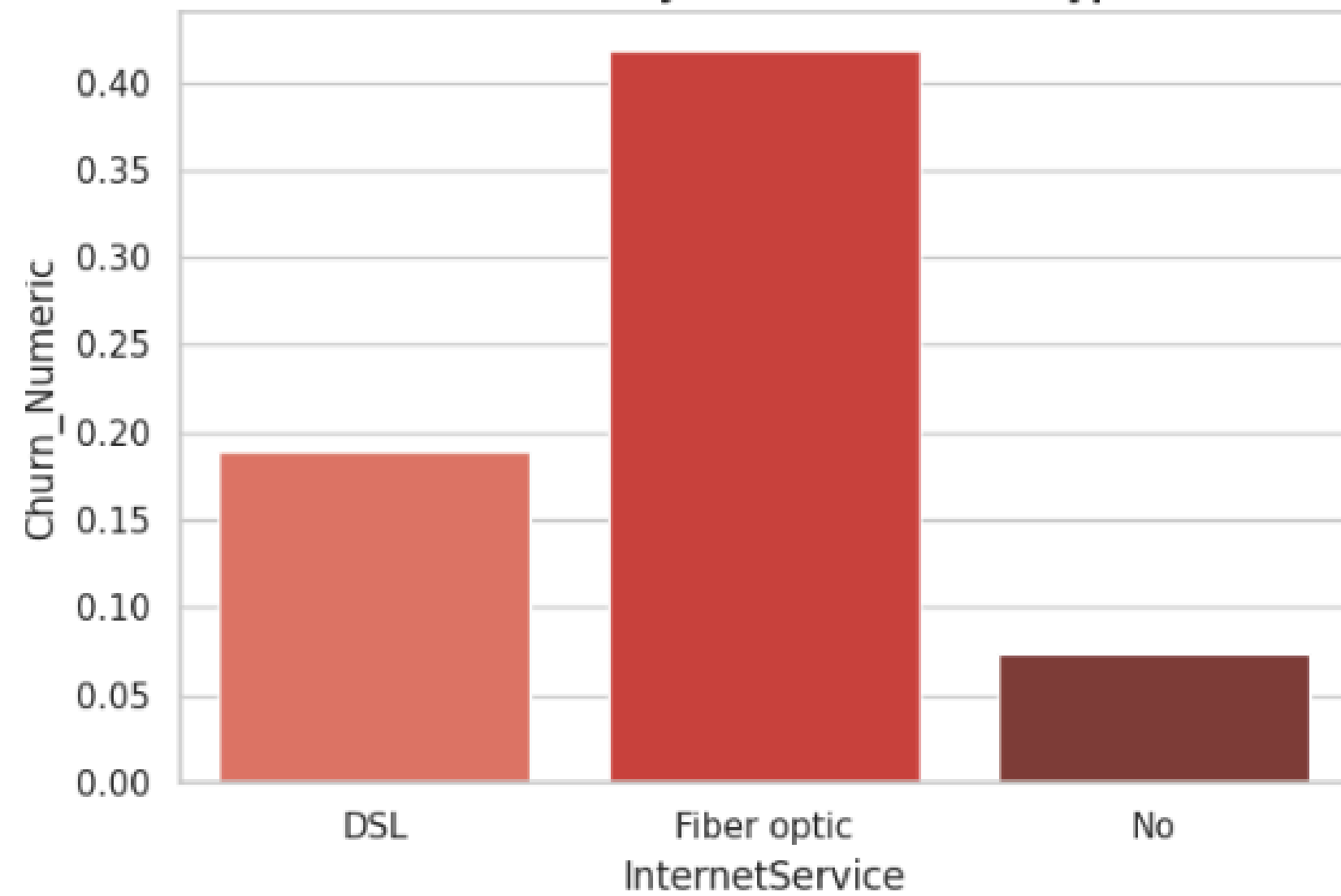
Churn Rate by Senior Citizen Status



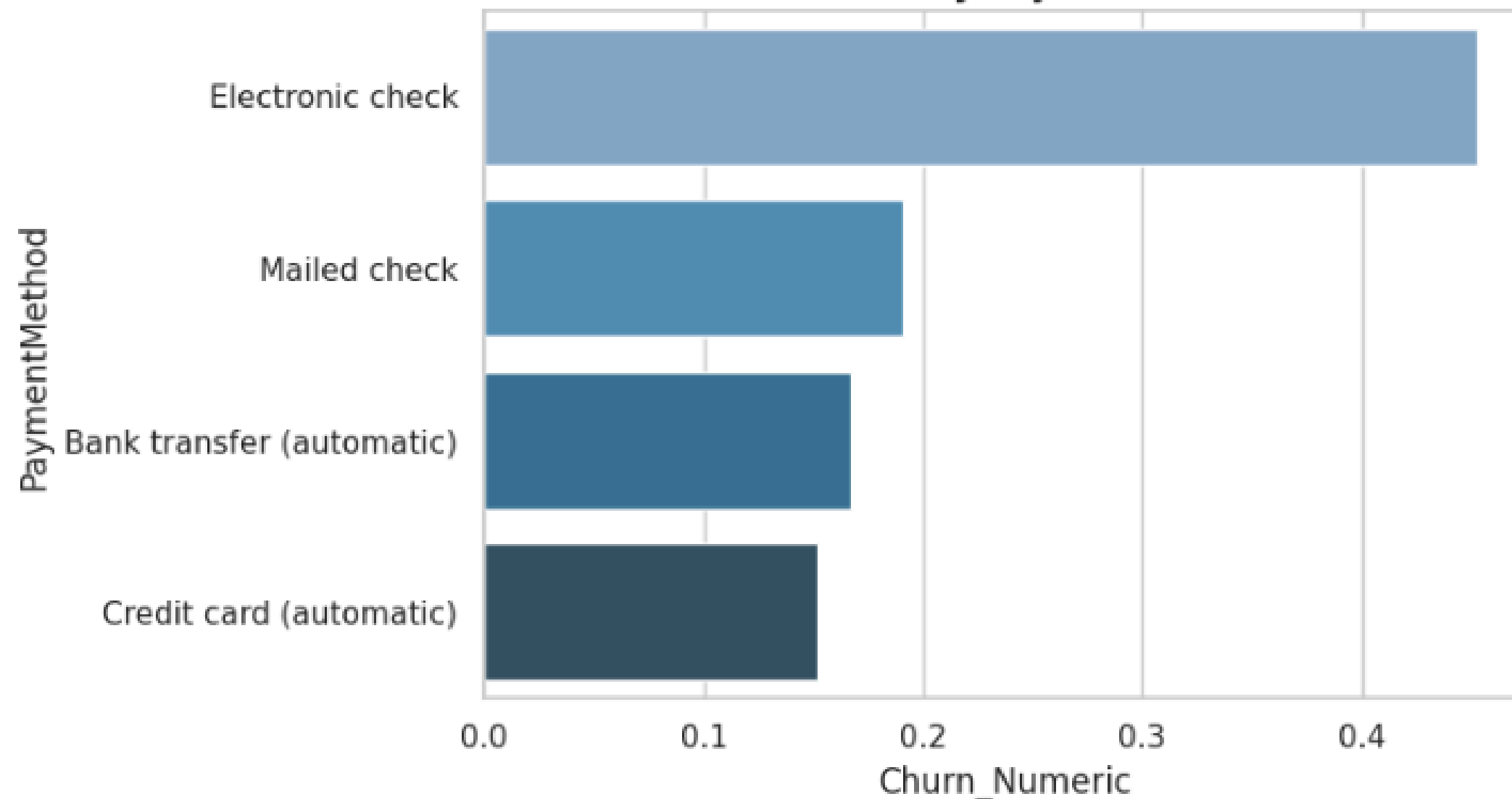
Churn Rate by Partner & Dependents Status



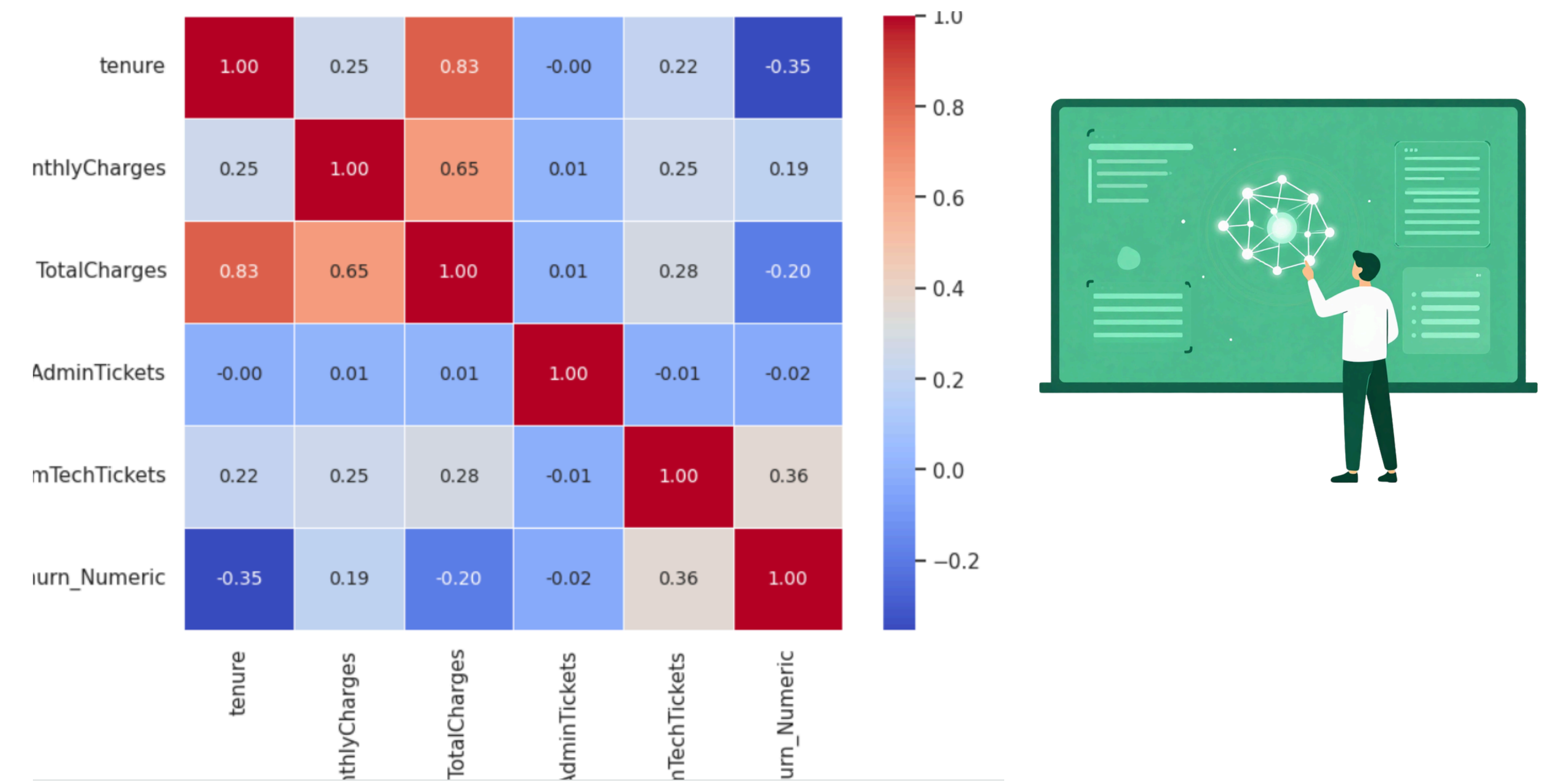
Churn Rate by Internet Service Type



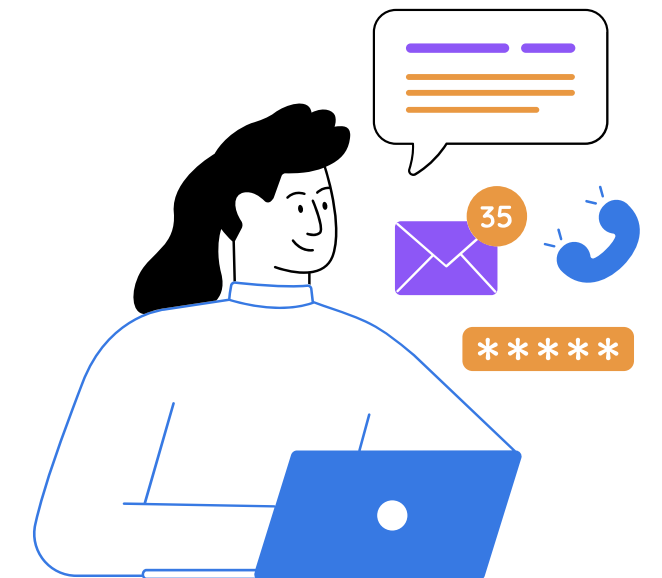
Churn Rate by Payment Method



Numerical Feature Correlation Heatmap



- KPI Summary
- Customer Base & Loss: Out of 7,043 total customers, 1,869 have churned, resulting in a high overall churn rate of 26.54%.
- Revenue Anchor: Across the entire customer dataset, the average monthly charge sits at \$64.76.
- Key Drivers & Risk Factors
- Contract Commitment: Customers on Month-to-Month contracts exhibit a drastically higher churn proportion compared to those on One-Year or Two-Year agreements. Flexible, short-term plans are the primary driver of customer drop-off.
- Tech Tickets & Friction: Churn volume surges among customers with higher counts of technical support tickets (1–3+ tickets), indicating that product or service issues directly accelerate churn.
- Price Sensitivity: Churned users show a higher concentration around elevated monthly charge distributions (especially the \$70–\$100 range), signaling price resistance among uncommitted tiers.
- Tenure Vulnerability: Customer loss is heavily front-loaded; the 0–1 year tenure cohort represents the largest absolute count of churned users, whereas retention stabilizes significantly after 2+ years.



- **Strategic Recommendations**
- Incentivize Contract Migration: Offer short-term discounts or bundled add-ons to migrate high-risk Month-to-Month subscribers onto long-term commitments.
- Enhance Early Onboarding: Deploy targeted retention offers during the critical first 12 months to improve tenure progression.
- Proactive Support Resolution: Flag accounts raising multiple tech tickets for priority support, preventing service frustration from turning into cancellations.



Thank You

~ANKITA SARKAR